



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

REVIEW TEST

06-06-2021

JEE(Main + Advance) : Enthusiast (PHASE : S,I(A),I, PHASE-I LIVE & PHASE I(A)

Test Type : REVIEW TEST

Test Pattern : JEE (Advanced)

TEST DATE : 06-06-2021

ANSWER KEY

PAPER-1

PHYSICS

SEC-1	Q.	1	2	3	4				
	A.	D	C	B	D				
SEC-2	Q.	1	2	3	4	5	6	7	8
	A.	A,B,C	A,B,D	A,D	B,C,D	A,B,C	B,C	A,B,D	B,C
SEC-3	Q.	1	2	3	4	5	6		
	A.	2.00	1.25	3.00	30.00	2.66 to 2.67	17.50		

CHEMISTRY

SEC-1	Q.	1	2	3	4				
	A.	B	A	A	D				
SEC-2	Q.	1	2	3	4	5	6	7	8
	A.	A,B,C,D	B,C	A,B	A,B,C	A,B,C,D	A,C,D	A,C	A,B,C,D
SEC-3	Q.	1	2	3	4	5	6		
	A.	375.00	7.00	8.00	4.00	2.00	5.00		

MATHEMATICS

SEC-1	Q.	1	2	3	4				
	A.	B	D	A	A				
SEC-2	Q.	1	2	3	4	5	6	7	8
	A.	A,B,C	A,B	A,B,C	A,B,C,D	A,B,C,D	A,C,D	A,C,D	B,C
SEC-3	Q.	1	2	3	4	5	6		
	A.	5.00	2.57, 2.58	0.00	0.50	3.14, 3.15	6.00		

HINT – SHEET

PHYSICS

SEC-1

1. Ans (D)

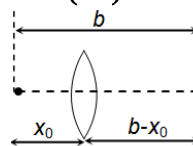
Direction of the frictional force is opposite to the direction of the velocity of the point of contact between the sphere and the ground.

∴ (d)

2. Ans (C)

The field inside cavity in this case is uniform so force on proton is constant. Therefore both straight line & parabola are possible.

3. Ans (B)



$m = -k$ (inverted real image)

$V = -ku$

$U = -x_0, v = b-x$

$b - x_0 = kx_0$

$$x_0 = \frac{b}{1+k}$$

Using lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$$\Rightarrow \frac{1}{b-x_0} - \frac{1}{-x_0} = P$$

$$\text{Putting } x_0 = \frac{b}{1+k}, P = \frac{(k+1)^2}{kb}$$

∴ (B)

4. Ans (D)

Let tangential velocity when radial distance becomes eR is V'

By conservation of angular momentum

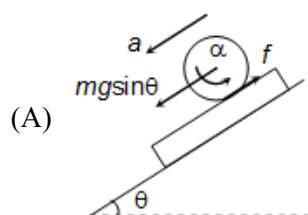
$$VR = V'eR$$

$$\text{Angular velocity} = \frac{V'}{eR} = \frac{V}{e^2R}$$

PHYSICS

SEC-2

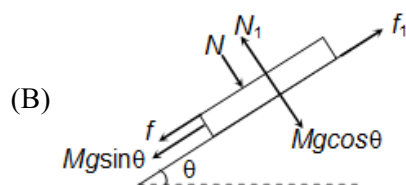
2. Ans (A,B,D)



When plank is at rest, for pure rolling

$$fr = \frac{Ia}{r} \text{ and } mg \sin \theta - f = ma$$

$$\Rightarrow a = \frac{5}{7}g \sin \theta \Rightarrow f = \frac{2}{7}mg \sin \theta$$

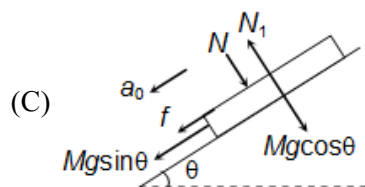


For $\mu_{\min}$

$$N_1 = (m + M)g \cos \theta$$

$$\frac{2}{7}mg \sin \theta + Mg \sin \theta = \mu N_1$$

$$\mu = \frac{(2m + 7M)}{7(M + m)} \tan \theta$$



$$Mg \sin \theta + f = Ma_0$$

$$mg \sin \theta - f = ma$$

$$fr = \frac{2}{5}mr^2\alpha$$

$$a_0 + \alpha r = a$$

$$\frac{f + Mg \sin \theta}{M} + \frac{5f}{2m} = \frac{mg \sin \theta - f}{m}$$

$$\Rightarrow f = 0$$

$\therefore$ (A), (B) and (D)

3. Ans (A,D)

Net field inside the conducting shell is zero

$$\text{Therefore, } \vec{E}_{\text{Outer charge}} + \vec{E}_{\text{induced charge}} = 0$$

$$\therefore |\vec{E}_{\text{induced}}| = \frac{kq}{d^2}$$

$$|\vec{E}_{\text{outer charge}}| = \frac{kq}{d^2}$$

$\therefore$ (A) and (D)

4. Ans (B,C,D)

$$U = U_0(1 - \cos ax),$$

$$F = \frac{-dU}{dx} = -aU_0 \sin ax$$

$$m \frac{d^2x}{dt^2} = -a^2 U_0 x \text{ (For small oscillations)}$$

$$\Rightarrow T = 2\pi \sqrt{\frac{m}{a^2 U_0}}$$

Potential energy is maximum at $x = \frac{\pi}{a}$

amplitude is $\frac{\pi}{a}$

Speed of particle is maximum at $x = 0$

($\because$ PE is minimum) (b), (c) and (d)

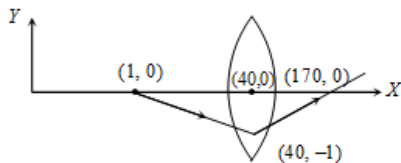
5. Ans (A,B,C)

$$\frac{1}{f} = (1.5 - 1) \left(\frac{1}{10} - \frac{1}{50} \right), f = 25 \text{ cm}$$

$$-\frac{1}{F} = \frac{2}{f} + \frac{1}{f_m} = \frac{2}{25} + \frac{1}{-25}, F = -25$$

$\therefore$ (A), (B) and (C)

6. Ans (B,C)



The ray is intersecting $y = 0$ line at $x = 1$ and $x = 40$ line at $y = -1$.

$$\therefore u = 39 \text{ cm}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow v = 130 \text{ cm}$$

$$\therefore \text{Equation of refracted ray is } 130y = x - 170$$

If space on the right of the lens is filled with liquid of $\mu = 4/3$, then

$$\frac{1.5}{v_1} + \frac{1}{39} = \frac{0.5}{30}$$

$$\frac{4}{3v} - \frac{1.5}{v_1} = \frac{(\frac{4}{3} - 1.5)}{-30}$$

$$v = -390 \text{ cm}$$

$\therefore$ Equation of refracted ray is

$\therefore$ (B) and (C)

7. Ans (A,B,D)

$$g = \frac{GM}{R^2} = \frac{G(\frac{4}{3}\pi R^3 \rho)}{R^2}, g \propto R,$$

$$\frac{g_1}{g_2} = \frac{R_1}{R_2} = 2$$

$$\frac{g_1}{g_3} = \frac{R_1}{R_2} = 3, v = \sqrt{2gR} \propto \sqrt{R^2} \propto R,$$

$$\frac{v_1}{v_2} = \frac{R_1}{R_2} = 2$$

$$\frac{v_1}{v_3} = \frac{R_1}{R_3} = 3$$

8. Ans (B,C)

$$v^2 = a - bx^2$$

differentiating both side w.r.t. time

$$2v \frac{dv}{dt} = -2bx \left(\frac{dx}{dt} \right)$$

$$\frac{dv}{dt} = A \text{ (acceleration)}$$

$$A = -bx$$

Hence motion will be S.H.M $\omega^2 = b$.

$$4\pi^2 f^2 = b, f = \frac{\sqrt{b}}{2\pi}$$

$$T.E. = \frac{1}{2} m \omega^2 (\text{Amplitude})^2 = \frac{1}{2} m b \frac{a}{b} = \frac{1}{2} m a$$

$\therefore$ (B, C)

PHYSICS

SEC-3

1. Ans (2.00)

$$PE = -4 \text{ MJ}$$

$$TE = -2 \text{ MJ}$$

The additional energy required to make the satellite escape = +2MJ.

2. Ans (1.25)

$$\vec{\tau} = \vec{p} \times \vec{E}$$

$$\tau_{\max} = p E$$

$$\Rightarrow 50 \times 10^{-28} = p.40$$

$$\Rightarrow p = 1.25 \times 10^{-28} \text{ C-m}$$

3. Ans (3.00)

$$\text{Force due to sheet} = \frac{\sigma}{2\epsilon_0} \times q$$

$$= \frac{10^{-9}}{72\pi\epsilon_0} 6 \times 10^{-3}$$

$$= 9 \times 10^9 \times \frac{6}{18} \times 10^{-3} \times 10^{-9} = 3 \text{ mN}$$

4. Ans (30.00)

$$V_B - V_A = - \int E_x dx = [\text{Area under } E_x - x \text{ curve}]$$

$$V_B - 10 = -\frac{1}{2} \cdot 2 \cdot (-20) = 20$$

$$V_B = 30 \text{ V}$$

5. Ans (2.66 to 2.67)

$$(\text{K.E.}) \text{ of plate w.r.to A} = \frac{1}{2} I_A \omega^2$$

$$I_A = \frac{4}{3} \text{ kg m}^2$$

$$(\text{K.E.})_A = \frac{1}{2} \times \frac{4}{3} \times 4 = \frac{8}{3} \text{ J}$$

6. Ans (17.50)

For Ist surface :

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{3/2}{v} - \frac{1}{\infty} = \frac{3/2 - 1}{+10} = \frac{1}{20}$$

$$v = 30 \text{ cm}$$

This will be the distance of object for second surface.

For IInd surface :

$$\frac{4/3}{v'} - \frac{3/2}{+30} = \frac{4/3 - 3/2}{-10}$$

$$\frac{4}{3v'} - \frac{1}{20} = \frac{1}{60}$$

$$v' = +20\text{cm.}$$

For IIIrd surface :

$$u = 20 - 10 = 10 \quad R = \infty$$

$$\frac{1}{V''} - \frac{4/3}{+10} = \frac{\mu_2 - \mu_1}{\infty} = 0$$

$$\frac{1}{V''} - \frac{4}{30} \Rightarrow V'' = \frac{30}{4}$$

The distance of final image from curved surface

$$10 + \frac{30}{4} = \frac{70}{4} = 17.5\text{cm.}$$

CHEMISTRY

SEC-1

1. **Ans (B)**

$$\frac{t_{\text{effuse CH}_4}}{t_{\text{effuse He}}} = \sqrt{\frac{m_{\text{CH}_4}}{m_{\text{He}}}}$$

$$\frac{48}{T} = \sqrt{\frac{16}{4}}$$

3. Ans (A)

Leaving ability of any leaving group $\propto$ size of atom

 $\propto EN$

$\propto$ delocalisation of $-ve$ charge

 $\propto e^{\ominus}$ withdrawing effect

Note : good leaving groups are weak bases

$$(A) \text{I}^{\ominus} > \text{Br}^{\ominus} > \text{Cl}^{\ominus} > \text{F}^{\ominus}(\text{size})$$
$$(B) \text{OH}^{\ominus} > \text{Et} - \text{O}^{\ominus} > \text{Me} - \overset{\text{O}}{\parallel} \text{C} - \text{O}^{\ominus}$$

Correct order :

$$\text{Me}-\overset{\text{O}}{\parallel}{\text{C}}-\text{O}^{\ominus} > \text{OH}^{\ominus} > \text{Et}-\text{O}^{\ominus}$$

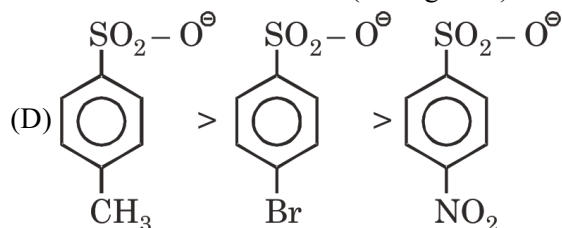
(Resonance)

stabilize)

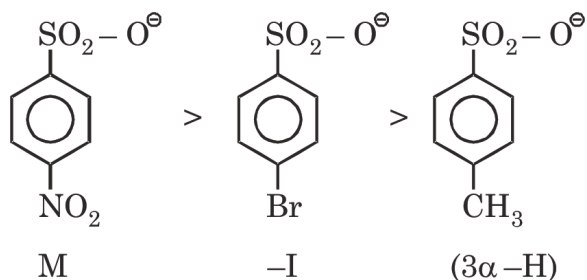
(C) $\text{NH}_2^- > \text{NMe}_3 > \text{NH}_3$

Correct order : $\text{NH}_3 > \text{NMe}_3 > \text{NH}_2^\ominus$

(Strong base)



Correct order :



CHEMISTRY

SEC-2

1. Ans (A,B,C,D)

$$\text{Distance between nearest neighbour} = \frac{\sqrt{3}a}{2}$$

(= 8 in number). Distance between next nearest neighbour = a (= 6 in number)

MATHEMATICS

SEC-1

1. **Ans (B)**

Apply L.M.V.T to $f(x)$ in $[1,3]$

$$f'(x) = \frac{f(3) - f(1)}{3 - 1}$$

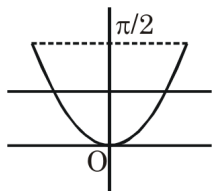
$$\Rightarrow \frac{a^2 - a}{2} \leq 6$$

$$\Rightarrow -3 \leq a \leq 4$$

2. **Ans (D)**

$$y = \sin^{-1}(|x|)$$

$$0 \leq \sin^{-1}(|x|) \leq \frac{\pi}{2}$$



for two solutions

$$0 < 4\sin^{-1}(|x|) \leq 2\pi$$

$$0 < 4\sin^{-1}(|x|) \leq 6.28$$

No. of integers = 6

3. **Ans (A)**

$$\log_2 \sin x - \log_2 \cos x - \log_2 (1 - \tan^2 x) = -1$$

Domain :

$$\left. \begin{array}{l} \sin x > 0 \\ \cos x > 0 \\ 1 - \tan^2 x > 0 \end{array} \right\}$$

$$x \in \left(2n\pi, 2n\pi + \frac{\pi}{4}\right)$$

$$\log_2 \left(\frac{\tan x}{1 - \tan^2 x} \right) = -1 \Rightarrow \frac{2 \tan x}{1 - \tan^2 x} = 1$$

$$\tan 2x = 1$$

$$2x = n\pi + \frac{\pi}{4} \Rightarrow x = \frac{n\pi}{2} + \frac{\pi}{8}$$

$$\text{Intersection with Domain : } x = \frac{\pi}{8}$$

$$\text{as } x \in [0, 2\pi)$$

4. **Ans (A)**

$$f(0) = \begin{vmatrix} 0 & 1 & 1 \\ 1 & 2 & 2 \\ 2 & 3 & 3 \end{vmatrix} = 0$$

$$\lim_{x \rightarrow 0} \frac{f(x)}{x^2} = \lim_{x \rightarrow 0} \frac{f'(x)}{2x} \quad \left(\frac{0}{0} \text{ form} \right)$$

$$f'(x) = \begin{vmatrix} 1 & \cos x & 1 \\ 2 & e^x - x + 1 & 2 \\ 3 & \sec x + 2 & 3 \end{vmatrix} + \begin{vmatrix} x & -\sin x & 1 \\ 2x+1 & e^x - 1 & 2 \\ 3x+2 & \sec x \tan x & 3 \end{vmatrix}$$

$$\frac{f'(x)}{x} = \begin{vmatrix} x & \left(-\frac{\sin x}{x}\right) & 1 \\ 2x+1 & \left(\frac{e^x-1}{x}\right) & 2 \\ 3x+2 & \left(\frac{\sec x \tan x}{x}\right) & 3 \end{vmatrix}$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{f'(x)}{2x} = \frac{1}{2} \begin{vmatrix} 0 & -1 & 1 \\ 1 & 1 & 2 \\ 2 & 1 & 3 \end{vmatrix}$$

$$= \frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ 1 & 3 & 2 \\ 2 & 4 & 3 \end{vmatrix} \quad C_2 \rightarrow C_2 + C_3$$

$$= \frac{1}{2} (4 - 6) = -1$$

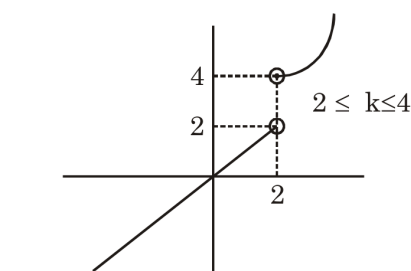
MATHEMATICS

SEC-2

1. **Ans (A,B,C)**

$$f(x) = x \quad \uparrow \quad \text{for } x < 2$$

$$f(x) = x^2 - 2x + 4 \quad \uparrow \quad \text{for } x > 2$$



2. Ans (A,B)

$$y = \frac{(a+x)^2}{x} \Rightarrow y = \frac{a^2}{x} + 2a + x$$

$$\frac{dy}{dx} = \frac{-a^2}{x^2} + 1 = -1 \text{ (for equal intercepts)}$$

$$x^2 = \frac{a^2}{2} \Rightarrow x = \pm \frac{a}{\sqrt{2}}$$

3. Ans (A,B,C)

$$f(x) = \tan^{-1}(\ln x)$$

$$f'(x) = \frac{1}{1 + \ln^2 x} \cdot \frac{1}{x}$$

and

$$f''(x) = -\frac{1}{(1 + \ln^2 x)^2 x^2} [(1 + \ln^2 x) + 2 \ln x]$$

$$g'(0) = \frac{1}{f'(1)} = 1$$

$$\text{and } g''(0) = -\frac{f''(1)}{(f'(1))^3} = 1$$

4. Ans (A,B,C,D)

$$\sin 18^\circ = \frac{\sqrt{5}-1}{4}; \cos 36^\circ = \frac{\sqrt{5}+1}{4}$$

$$\tan \frac{3\pi}{8} = \sqrt{2} + 1; \tan \frac{\pi}{8} = \sqrt{2} - 1$$

$$\cot 75^\circ = 2 - \sqrt{3}; \cos 15^\circ = 2 + \sqrt{3}$$

$$\log \left(\frac{\sqrt{5}}{2} \cdot 2.4^{\frac{1}{4}} \right) = \log_{10} \left(10^{\frac{1}{2}} \right) = \frac{1}{2}$$

$$(A) \sin 1050^\circ = -\frac{1}{2}$$

$$(B) \cos 150^\circ = 2$$

$$(C) \sin 54^\circ - \cos 72^\circ$$

$$= \frac{\sqrt{5}+1}{4} - \frac{\sqrt{5}-1}{4} = \frac{1}{2}$$

$$(D) x = \sqrt{12+x} \Rightarrow x^2 - x - 12 = 0 \Rightarrow x = 4$$

$$\log_4 2 = \frac{1}{2}$$

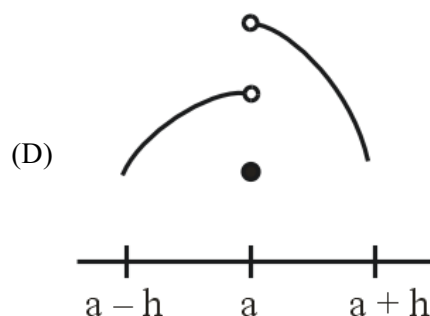
5. Ans (A,B,C,D)

(A) $f(x) = x^{1/3}$ vertical tangent at $x = 0$ but $f(x)$ is non-derivable

$$(B) f(x) = \begin{cases} x^2 \sin \frac{1}{x} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

$f(x)$ is derivable but $f'(x)$ is discontinuous

(C) Statement is true when $f(x)$ is continuous in $[a,b]$



inc in $(a-h, a)$

dec in $(a, a+h)$

but not maximum at $x = a$

6. Ans (A,C,D)

$$(x) = \frac{2021^{2x} - 1}{2021^{2x} + 1} = 1 - \frac{2}{2021^{2x} + 1}$$

$\Rightarrow$ range of x is $(-1, 1)$ ($\because 2021^{2x} > 0$)

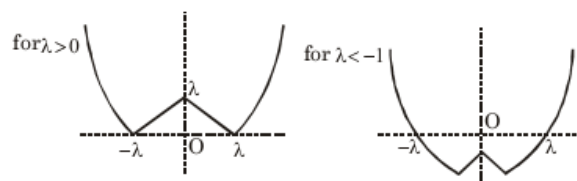
Domain : $2021^{2x} + 1 \neq 0 \Rightarrow x \in \mathbb{R}$

$$\text{Now, } f(-x) = \frac{2021^{-2x} - 1}{2021^{-2x} + 1}$$

$$= \frac{1 - 2021^{2x}}{1 + 2021^{2x}} = -f(x) \Rightarrow f(x) \text{ is odd}$$

function

7. Ans (A,C,D)



for $-1 \leq \lambda \leq 0$ $f(x) = x^2 - \lambda^2$ which is always differentiable

8. Ans (B,C)

$$\frac{\log_8 \left(\frac{8}{x^2} \right)}{(\log_8 x)^2} = 3$$

$$\frac{1 - 2\log_8 x}{(\log_8 x)^2} = 3$$

Let $\log_8 x = t$

$$1 - 2t = 3t^2$$

$$\Rightarrow 3t^2 + 2t - 1 = 0$$

$$\Rightarrow (3t + 1)(t - 1) = 0$$

$$\Rightarrow t = -\frac{1}{3}, t = 1$$

$$\log_8 x = \frac{1}{3}, \log_8 x = -1$$

$$\Rightarrow \boxed{x = 2} \quad \boxed{x = \frac{1}{8}}$$

MATHEMATICS

SEC-3

1. Ans (5.00)

$f(x)$ has maximum value 8 which occurs when $\cos^2 x = 1$

$$\Rightarrow a + b = 8$$

similarly $f(x)$ has minimum value -3 which occurs when $\cos^2 x = 0$

$$\Rightarrow b = -3$$

$$\Rightarrow a + 2b = 5$$

2. Ans (2.57, 2.58)

$$f(-1) = 0; \quad g(-1) = 0;$$

$$f'(-1) = g'(-1)$$

$$\Rightarrow a = -1; \quad b = -1; \quad c = 1;$$

$$\text{hence } |\sin^{-1}(-1) + \sin^{-1}(\sin(-1)) + \cos^{-1}(1)|$$

$$\left| -\frac{\pi}{2} - 1 \right| = \frac{\pi}{2} + 1 = 2.57$$

3. Ans (0.00)

$$x^2 y^2 = 81 \Rightarrow 2y \frac{dy}{dx} = \frac{-162}{x^3}$$

$$\Rightarrow \alpha^2 \beta^2 = 81$$

$$\text{length of sub tangent} = \left| \frac{\beta}{(dy/dx)_{\alpha, \beta}} \right|$$

$$\Rightarrow \left| \frac{\beta \cdot \alpha^3 \cdot \beta}{-81} \right| = \left| \frac{81\alpha^3}{\alpha^2 \cdot -81} \right| = 3$$

$$\Rightarrow \alpha = \pm 3 \Rightarrow \beta = \pm 3$$

Pts (3,3), (3,-3), (-3,3), (-3,-3)

$$\alpha + \beta = 6 \quad 0 \quad 0 \quad -6$$

$$\Sigma(\alpha + \beta) = 0$$

4. Ans (0.50)

$$\frac{dy}{dx} = \frac{k \cos(kt)}{\cos t} = y_1$$

$$(1 - \sin^2 t) y_1^2 = k^2 (1 - \sin^2 kt)$$

$$(1 - x^2) y_1^2 = k^2 (1 - y^2)$$

Differentiate both the sides

$$(1 - x^2)^2 y_1 y_2 - 2x^2 = -2k^2 y y_1$$

$$(1 - x^2) y_2 - x y_1 + k^2 y = 0$$

$$\lambda_1 = 1, \lambda_2 = 1$$

$$\frac{\lambda_1}{2\lambda_2} = 0.50$$

5. Ans (3.14, 3.15)

$$f(x) = \sqrt{\cos^{-1}(2x) + 2} + \sqrt{1 - \cos^{-1}(2x)}$$

$$1 - \cos^{-1}(2x) \geq 0$$

$$\Rightarrow -1 \leq 2x \leq 1$$

$$\cos^{-1}(2x) \leq 1$$

$$\Rightarrow -\frac{1}{2} \leq x \leq \frac{1}{2}$$

$$2x \geq \cos(1)$$

$$x \geq \frac{1}{2} \cos 1$$

Domain is

$$\frac{1}{2} \cos 1 \leq x \leq \frac{1}{2}$$

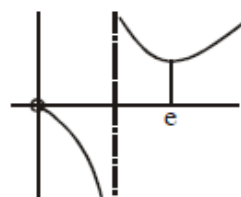
$$\& \text{ Range : } [\sqrt{3}, \sqrt{2} + 1]$$

$$a + b = 3.146$$

6. Ans (6.00)

$$f'(x) = \frac{\log x - 1}{(\log x)^2}$$

$f(x)$ increasing in $x > e$ & decreasing in $0 < x < e$
discontinuous at $x = 1$



$$\therefore a \in (1, e) \Rightarrow a = 2$$

$$f(a) = (b) \Rightarrow \frac{2}{\log 2} = \frac{b}{\log b}$$

$$b \in \mathbb{N} \quad \therefore b = 2^k$$

$$\frac{2}{\log 2} = \frac{2^k}{k \log 2} \Rightarrow 2^k = 2k$$

$$k = 2 \quad \therefore b = 4$$

$$a + b = 6$$